EXAM PACK

EEE 0714 3229

Communication Engineering I



Exam Pack

Session 2022–23 *Solutions and formula sheet of Term Tests 2 and 3 (the final will be added)*

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Prepared by Nur Mohammad

November 2026

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PAST PAPER SOLUTIONS

EEE 0714 3229

Communication Engineering I

Question Paper Solutions

Session 2022–23

Term Tests 2 and 3 (Q7–Q8); the final examination will be added as Q1–Q6

QUESTION MAP

Q7	Q8
–	20 marks

EACH ANSWER

Theory → Answer → Keyword Explanation → Notes and Remarks → Final Concise Answer

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November 2026

BACK OF THE COVER

The paper at a glance

Q	Sub-part	Marks	Time	Pick
Term Test 2 (its questions 1 to 10 are 7(a) to 7(j)) · Communication – TT2 Questions				
Q7	(a) Advantages and Disadvantages of FM over AM			☐
	(b) Modulation Index, Sidebands, Power and Efficiency of an AM Signal			
	(c) Switching Modulators: the Ring Modulator			
	(d) Comparison of AM (DSB-WC), DSB-SC and SSB-SC			
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Term Test 3 · 20 marks · 30 min				
Q8	(a) Design of an Armstrong FM Modulator for 97.3 MHz and 20.48 kHz	10	15 min	☐
	(b) DPSK Modulator, Demodulator and the Sequence 1010011100	4	6 min	
	(c) Pulse-Stuffing TDM of 190 kbps and 180 kbps Channels	6	9 min	

Formulas and conventions follow Lathi and Ding, Modern Digital and Analog Communication Systems (4th ed.); 8(c) follows Forouzan, Data Communications and Networking.

Shahjalal University of Science and Technology, Sylhet
Department of Electrical and Electronic Engineering
EEE 0714 3229 Communication Engineering 1
Session: 2022-23

Term Tests 2 and 3 **Session: 2022-23** **Term Test 3 held on 10/11/26**

Term Test 2 (its questions 1 to 10 are 7(a) to 7(j))

[Communication – TT2 Questions]

7. (a) Advantages and disadvantages of FM over AM.
   (b) An audio frequency signal $10\sin 2\pi 500t$ is used to amplitude modulate a carrier of $50\sin 2\pi 10^5 t$. Calculate:
      (i) Modulation index
      (ii) Sideband frequencies
      (iii) Amplitude of each sideband frequency
      (iv) Bandwidth required
      (v) Total power delivered to the load of 600 Ω
      (vi) Transmission efficiency
   (c) Explain switching modulators, specifically the ring modulator: circuit, working, and the case $m(t) = 0$ where the carrier is suppressed.
   (d) Compare AM (DSB-WC), DSB-SC and SSB-SC in modulation, demodulation, power efficiency and bandwidth.
   (e) Explain the effect of local oscillator phase error and frequency error in coherent detection of DSB-SC/SSB.
   (f) Disadvantages of Weaver's method of SSB generation.
   (g) What is VSB (vestigial sideband)? Why is it used?
   (h) Why is VSB-WC chosen for TV broadcasting?
   (i) Derive the narrowband FM ($\beta < 1$) expression.
   (j) Draw the NBFM block diagram.

Term Test 3 (its questions 1 to 3 are 8(a) to 8(c))

[Term Test 3, Time: 30min, Total Marks: 20]

8. (a) Design an Armstrong indirect FM modulator to generate an FM signal with carrier frequency 97.3MHz and $\Delta f = 20.48kHz$. A NBFM generator of $f_{c1} = 20kHz$ and $\Delta f = 5Hz$ is available. Only frequency doublers can be used as multipliers. Additionally, a local oscillator (LO) with adjustable frequency between 400 and 50-kHz [*sic*: should be "500 kHz"] is readily available for frequency mixing. 10
   (b) Draw the block diagram of a Differential PSK (DPSK) modulator and demodulator and show how the I_k sequence [1010011100] changes in different parts of the block. 4
   (c) Two channels, one with a bit rate of 190 kbps and another with a bit rate of 180 kbps, are to be multiplexed using pulse stuffing TDM with no synchronization bits. Answer the following questions:
      a. What is the size of a frame in bits?

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b. What is the frame rate?
c. What is the duration of a frame?
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7. (a) Advantages and disadvantages of FM over AM.
   (b) An audio frequency signal $10\sin 2\pi 500t$ is used to amplitude modulate a carrier of $50\sin 2\pi 10^5 t$. Calculate:
      (i) Modulation index
      (ii) Sideband frequencies
      (iii) Amplitude of each sideband frequency
      (iv) Bandwidth required
      (v) Total power delivered to the load of 600 Ω
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   (c) Explain switching modulators, specifically the ring modulator: circuit, working, and the case $m(t) = 0$ where the carrier is suppressed.
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   (f) Disadvantages of Weaver's method of SSB generation.
   (g) What is VSB (vestigial sideband)? Why is it used?
   (h) Why is VSB-WC chosen for TV broadcasting?
   (i) Derive the narrowband FM ($\beta < 1$) expression.
   (j) Draw the NBFM block diagram.

7.(a) Advantages and disadvantages of FM over AM.

1. Theory

1. Where the message lives (Core Idea)

In **AM** the message changes the carrier's **amplitude**; in **FM** it changes the carrier's **frequency**, and the amplitude stays constant. Almost all noise and interference on a radio channel adds to the **amplitude** of the received signal. So in AM the noise lands directly on the message, while in FM the receiver can throw the amplitude away (with a limiter) and keep only the frequency, where the noise does much less harm.

2. The price: bandwidth

An AM signal needs twice the message bandwidth, $2B$. An FM signal needs about $2(\Delta f + B)$ by Carson's rule, usually several times more. FM trades bandwidth for noise performance.

2. Answer

Advantages of FM over AM:

1. **Better noise immunity:** amplitude noise is removed by the limiter; the output SNR of wideband FM is higher than AM's by about $3\beta^2$ (for tone modulation), and it grows with the bandwidth used (bandwidth–SNR trade-off).
2. **Constant envelope:** the transmitter can use efficient nonlinear (class C) power amplifiers; all transmitted power is useful.
3. **Capture effect:** of two FM stations on the same frequency, the receiver locks onto the stronger one and suppresses the weaker, so co-channel interference is small.
4. **Higher fidelity:** FM broadcasting carries audio up to 15 kHz (AM about 5 kHz), and pre-emphasis/de-emphasis further reduce noise.
5. **Less adjacent-channel interference,** thanks to guard bands in the FM broadcast band.

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Disadvantages of FM compared with AM:

1. **Much larger bandwidth:** about 200 kHz per broadcast channel, against 10 kHz for AM; so FM is placed at VHF (88–108 MHz).
2. **Shorter range:** at VHF the waves travel only by line of sight (about 50–80 km), while AM medium-wave signals travel far by ground wave.
3. **More complex and costly** transmitters and receivers (limiter, discriminator or PLL, de-emphasis).
4. **Threshold effect:** below a certain carrier-to-noise ratio the FM output breaks down suddenly.

Final Summary: FM gives better noise performance, efficient constant-envelope transmission, the capture effect and higher fidelity, at the cost of much more bandwidth, shorter range and more complex equipment.

3. Notes and Remarks

- Lathi, Ch. 5 and Ch. 10 (the noise comparison of AM and FM).
- The Term Test 2 sheet ("Communication – TT2 Questions") prints no marks, so its ten items carry none here.

7.(b) An audio frequency signal $10\sin 2\pi 500t$ is used to amplitude modulate a carrier of $50\sin 2\pi 10^5 t$. Calculate: (i) Modulation index (ii) Sideband frequencies (iii) Amplitude of each sideband frequency (iv) Bandwidth required (v) Total power delivered to the load of 600 Ω (vi) Transmission efficiency

1. Theory

1. AM with a tone (Core Idea)

An AM signal is $[E_c + E_m \sin\omega_m t]\sin\omega_c t = E_c[1+\mu\sin\omega_m t]\sin\omega_c t$, where the **modulation index** is $\mu = E_m/E_c$. Using $\sin A\sin B = \frac{1}{2}[\cos(A-B)-\cos(A+B)]$:

$$\mu E_c \sin\omega_m t\sin\omega_c t = \frac{\mu E_c}{2}\cos(\omega_c-\omega_m)t - \frac{\mu E_c}{2}\cos(\omega_c+\omega_m)t$$

So there is a carrier of amplitude E_c and two side frequencies at $f_c \pm f_m$ of amplitude $\mu E_c/2$.

2. Power and efficiency

A sinusoid of peak E in a resistance R gives $E^2/2R$. So $P_c = E_c^2/2R$, each sideband $\mu^2 P_c/4$, and

$$P_t = P_c\left(1+\frac{\mu^2}{2}\right), \qquad \eta = \frac{\text{sideband power}}{P_t} = \frac{\mu^2}{2+\mu^2}$$

2. Answer

Given: $E_m = 10$ V, $f_m = 500$ Hz; $E_c = 50$ V, $f_c = 10^5$ Hz $= 100$ kHz; $R = 600\ \Omega$.

Step 1: Modulation index

$$\mu = \frac{E_m}{E_c} = \frac{10}{50} = 0.2 \quad (20\%)$$

Step 2: Sideband frequencies

$$f_{USB} = 100 + 0.5 = 100.5 \text{ kHz}, \qquad f_{LSB} = 100 - 0.5 = 99.5 \text{ kHz}$$

Step 3: Amplitude of each sideband

$$\frac{\mu E_c}{2} = \frac{0.2\times 50}{2} = 5 \text{ V}$$

Step 4: Bandwidth

$$B_T = 2f_m = 2 \times 500 = 1000 \text{ Hz} = 1 \text{ kHz}$$

Step 5: Total power in 600 Ω

$$P_c = \frac{E_c^2}{2R} = \frac{50^2}{2 \times 600} = \frac{2500}{1200} = 2.0833 \text{ W}$$

$$P_t = P_c \left(1 + \frac{\mu^2}{2}\right) = 2.0833 \times 1.02 = 2.125 \text{ W}$$

(each sideband carries $\mu^2 P_c/4 = 0.0208$ W).

Step 6: Transmission efficiency

$$\eta = \frac{\mu^2}{2 + \mu^2} = \frac{0.04}{2.04} = 0.0196 = 1.96\%$$

$$\boxed{\mu = 0.2;\ 100.5 \text{ \& } 99.5 \text{ kHz};\ 5 \text{ V};\ 1 \text{ kHz};\ 2.125 \text{ W};\ 1.96\%}$$

Final Answer: $\mu = 0.2$; sidebands at 100.5 kHz and 99.5 kHz, 5 V each; bandwidth 1 kHz; total power 2.125 W; efficiency 1.96 %.

3. Notes and Remarks

- The low efficiency shows why AM wastes power: at 20 % modulation only about 2 % of the power carries the message.

7.(c) Explain switching modulators, specifically the ring modulator: circuit, working, and the case $m(t) = 0$ where the carrier is suppressed.

1. Theory

1. Why switching? (Core Idea)

A DSB-SC modulator must multiply $m(t)$ by a carrier. An exact analog multiplier is hard to build at high power, but a **switch** is easy. If $m(t)$ is multiplied by a periodic switching function $w(t)$ of carrier frequency, $w(t)$ contains a $\cos \omega_c t$ term (Fourier series), so the product contains $m(t) \cos \omega_c t$; a bandpass filter keeps it and removes the rest. This is a **switching modulator**. The **ring modulator** is the version in which the switch reverses the polarity of $m(t)$: $w(t)$ is a square wave of ± 1.

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2. Answer

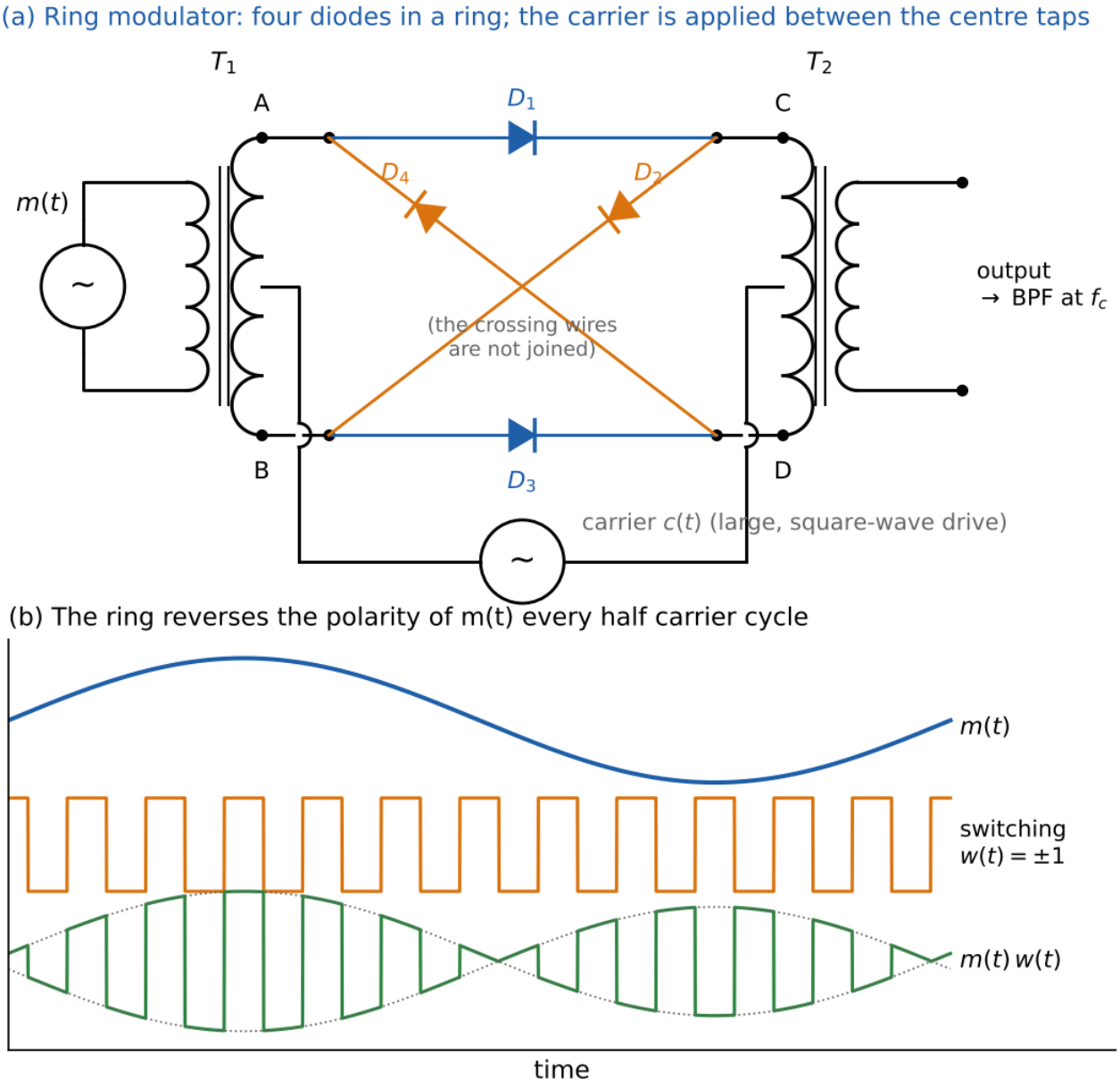


Figure 1: (a) The ring modulator circuit; (b) its action: m(t) is multiplied by a ±1 square wave

Reading the figure. In (a) the message enters the input transformer T_1; the ends A and B of its secondary connect, through the four diodes, to the ends C and D of the output transformer T_2. D_1 (A→C) and D_3 (B→D), in blue, form the outer pair; D_2 (C→B) and D_4 (D→A), in orange, cross; together the four point the same way round the ring. The carrier source is connected between the centre taps of T_1 and T_2, and the output goes to a bandpass filter. In (b) the blue message, the orange ±1 switching wave, and the green product, which jumps between $+m(t)$ and $-m(t)$ (dotted grey) every half carrier cycle.

Step 3 is marked **Basic** (the Fourier series of a square wave); a student who knows it can skip it.

Step 1: The circuit

Four diodes in a ring (each pointing to the next), between the input transformer T_1 (which carries $m(t)$) and the output transformer T_2. The carrier, much larger than $m(t)$, is applied between the two centre taps, so the carrier, not the message, decides which diodes conduct.

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Step 2: Working

- **Carrier positive half-cycle:** D_1 and D_3 conduct, D_2 and D_4 are off. A is joined to C and B to D: $m(t)$ passes straight through, output $+m(t)$.
- **Carrier negative half-cycle:** D_2 and D_4 conduct. A is joined to D and B to C: the connections are crossed, output $-m(t)$.

So the output is $m(t)w(t)$, where $w(t)$ is a square wave that switches between $+1$ and -1 at the carrier frequency.

Step 3 (Basic): Fourier series of the ± 1 square wave

A square wave of value $+1$ for $|\omega_c t| < \pi/2$ and -1 for $\pi/2 < |\omega_c t| < \pi$ is even, so only cosines appear, with coefficients $a_n = \frac{2}{\pi}\int_0^{\pi} w \cos n\theta \, d\theta = \frac{4}{n\pi}\sin\frac{n\pi}{2}$ (zero for even n):

$$w(t) = \frac{4}{\pi}\left[\cos\omega_c t - \frac{1}{3}\cos 3\omega_c t + \frac{1}{5}\cos 5\omega_c t - \cdots\right]$$

It has **no dc term**.

Step 4: The output

$$m(t)w(t) = \frac{4}{\pi}m(t)\cos\omega_c t - \frac{4}{3\pi}m(t)\cos 3\omega_c t + \cdots$$

A bandpass filter at f_c (bandwidth $2B$) keeps the first term:

$$\boxed{\varphi(t) = \frac{4}{\pi}m(t)\cos\omega_c t}$$

a DSB-SC signal. Because $w(t)$ has no dc term, there is no $m(t)$ term at baseband either: the ring is **double balanced**: neither the carrier nor the message appears at the output.

Step 5: The case $m(t) = 0$ (carrier suppressed)

With no message, the output is $0 \times w(t) = 0$. Physically, the carrier current enters T_2 at its centre tap and splits equally between the two halves of its primary (through D_1 and D_3, or D_2 and D_4). The two halves carry equal currents in opposite directions, so their magnetic fluxes cancel and nothing reaches the secondary. The carrier is therefore **suppressed**: the output exists only when there is a message, which is exactly DSB-SC.

Steps to remember

Large carrier switches the diode pairs $\to$ output $\pm m(t) = m(t)w(t) \to$ square wave $= \frac{4}{\pi}[\cos\omega_c t - \frac{1}{3}\cos 3\omega_c t + \ldots]$, no dc $\to$ BPF at f_c: $\frac{4}{\pi}m\cos\omega_c t \to m = 0$: balanced currents cancel, no carrier.

3. Notes and Remarks

- Lathi, Sec. 4.2 (switching and ring modulators).
- Unbalanced diodes or transformers let a little carrier leak through (carrier leakage).

7.(d) Compare AM (DSB-WC), DSB-SC and SSB-SC in modulation, demodulation, power efficiency and bandwidth.

1. Theory

1. Three members of the AM family (Core Idea)

All three put the message into the carrier's amplitude. They differ in what they send besides the message: **AM** sends a carrier and both sidebands; **DSB-SC** sends both sidebands but no carrier; **SSB-SC** sends only one sideband.

2. Answer

Feature	AM (DSB-WC)	DSB-SC	SSB-SC
Signal	$[A + m(t)] \cos \omega_c t$	$m(t) \cos \omega_c t$	$m \cos \omega_c t \mp m_h \sin \omega_c t$
Modulation	simple (nonlinear or switching modulator); needs $A \geq m_p$	balanced (product or ring) modulator	DSB-SC then a sharp sideband filter, or the phase-shift / Weaver method
Demodulation	envelope detector (cheap, no carrier recovery)	coherent detector (local carrier with exact frequency and phase; Costas loop)	coherent detector (or envelope detection after adding a large carrier)
Power efficiency	poor: at most 33.3 % (at $\mu = 1$); most power in the carrier	100 % of the power in the sidebands	100 %, and only one sideband is sent
Bandwidth	$2B$	$2B$	B
Effect of sync errors	none (envelope detection)	attenuation ($\cos \delta$), null at 90°, beating	phase distortion, frequency shift of the output
Use	broadcasting (one transmitter, many cheap receivers)	stereo FM subcarrier, QAM, analog data	point-to-point voice (radio, telephone FDM)

Table 1: Summary

Final Summary: AM is the simplest to receive but wastes power; DSB-SC saves the carrier power but needs coherent detection; SSB-SC also halves the bandwidth but needs the hardest modulator and receiver.

3. Notes and Remarks

- Lathi, Sec. 4.2 to 4.4.

7.(e) Explain the effect of local oscillator phase error and frequency error in coherent detection of DSB-SC/SSB.

1. Theory

1. Coherent detection (Core Idea)

The receiver multiplies the signal by a local carrier and low-pass filters it. The local oscillator should match the transmitter's carrier exactly, but in practice it is $2\cos[(\omega_c + \Delta\omega)t + \delta]$, with a small frequency error $\Delta\omega$ and phase error δ.

2. Answer

Step 1 is marked **Basic** (the product identities); a student who knows them can skip it.

Step 1 (Basic): Identities

$$2\cos A\cos B = \cos(A-B) + \cos(A+B), \qquad 2\sin A\cos B = \sin(A-B) + \sin(A+B)$$

Step 2: DSB-SC

$$m(t)\cos\omega_c t \cdot 2\cos[(\omega_c + \Delta\omega)t + \delta] \xrightarrow{\text{LPF}} e_o(t) = m(t)\cos(\Delta\omega t + \delta)$$

- **Phase error only:** $e_o = m(t)\cos\delta$: attenuation, zero at $\delta = 90°$ (quadrature null); a varying δ makes the output fade.
- **Frequency error only:** $e_o = m(t)\cos\Delta\omega t$: the output swells and fades at the beat rate Δf, a severe distortion.

Step 3: SSB (lower sideband) $\varphi = m(t)\cos\omega_c t + m_h(t)\sin\omega_c t$

Multiply each term by $2\cos[(\omega_c + \Delta\omega)t + \delta]$ and low-pass filter, using the identities with $A = \omega_c t$, $B = (\omega_c + \Delta\omega)t + \delta$ (so $A - B = -(\Delta\omega t + \delta)$):

$$e_o(t) = m(t)\cos(\Delta\omega t + \delta) - m_h(t)\sin(\Delta\omega t + \delta)$$

Step 4: SSB, phase error only

$$e_o(t) = m(t)\cos\delta - m_h(t)\sin\delta$$

For a tone $m = \cos\omega_m t$, $m_h = \sin\omega_m t$: $e_o = \cos(\omega_m t + \delta)$. Every component keeps its amplitude but is **phase-shifted by** δ: phase distortion. The ear hardly notices it, so it is acceptable for voice, but not for video or data.

Step 5: SSB, frequency error only

For the tone: $e_o = \cos\omega_m t\cos\Delta\omega t - \sin\omega_m t\sin\Delta\omega t = \cos(\omega_m + \Delta\omega)t$. Every component is **shifted in frequency by** Δf: the output is the message moved up or down in frequency. Speech sounds unnatural ("Donald Duck" voice) but stays intelligible for small Δf (a few tens of hertz).

Final Summary:

	DSB-SC	SSB
Phase error δ	attenuation by $\cos\delta$; null at 90°	each component phase-shifted by δ (phase distortion), no loss
Frequency error $\Delta\omega$	beating: $m(t)\cos\Delta\omega t$	every frequency shifted by Δf

Table 2: Summary

Steps to remember

Signal $\times\, 2\cos[(\omega_c+\Delta\omega)t+\delta] \to$ identities $\to$ LPF $\to$ DSB: $m\cos(\Delta\omega t+\delta)$; SSB: $m\cos(\cdot) - m_h\sin(\cdot)$ $\to$ tone: phase shift δ or frequency shift Δf.

3. Notes and Remarks

- Lathi, Sec. 4.2 and 4.4 (SSB demodulation with carrier errors).

7.(f) Disadvantages of Weaver's method of SSB generation.

1. Theory

1. Weaver's method (Core Idea)

Weaver's method makes SSB without a sharp sideband filter and without a wideband $-90°$ phase shifter. It uses two quadrature stages. The first multiplies the message by $\cos\omega_0 t$ and $\sin\omega_0 t$, where ω_0 is the **centre** of the message band, and low-pass filters both to half the bandwidth: this folds the message band around its centre. The second stage multiplies by $\cos\omega_2 t$ and $\sin\omega_2 t$ and adds (or subtracts) the two branches; the unwanted sideband cancels, as in the phase-shift method.

2. Answer

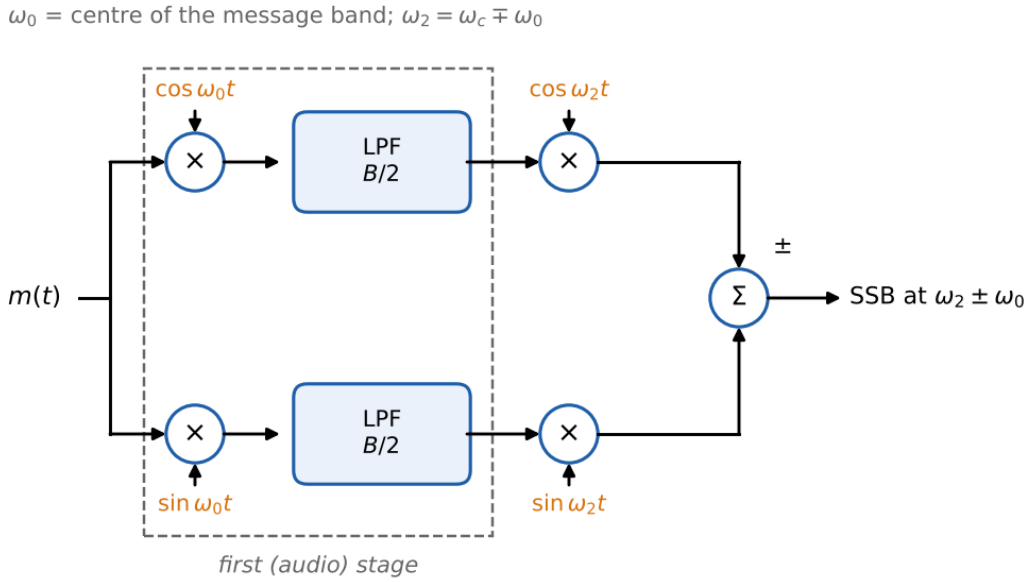


Figure 2: Weaver's SSB modulator: two quadrature stages with low-pass filters in between

Reading the figure. The message splits into two branches. Each is multiplied by an audio-frequency oscillator at the band centre (orange $\cos \omega_0 t$ above, $\sin \omega_0 t$ below), low-pass filtered to $B/2$ (the dashed box is this first, audio stage), then multiplied by a carrier pair ($\cos \omega_2 t$, $\sin \omega_2 t$). The summer adds or subtracts the two branches to give the SSB signal.

Disadvantages:

1. **Complexity:** four multipliers, two pairs of quadrature oscillators (audio and RF) and two low-pass filters, against one modulator and one filter in the filter method.
2. **Matching:** the two branches must have exactly equal gain and exactly 90° phase difference in *both* oscillator pairs, and the two LPFs must be identical. Any imbalance leaves part of the unwanted sideband.
3. **The leftover lands inside the band:** because the first stage folds the band about its centre, the unwanted residue is not outside the channel (as in the phase-shift method) but on top of the wanted signal, mirrored about ω_0; it cannot be filtered out and is heard as distortion.
4. **Tone at the band centre:** any dc offset or carrier leakage in the first stage turns into a tone at ω_0, right in the middle of the audio band (for speech, about 1.65 kHz), which is audible.
5. **Two carrier frequencies:** the RF oscillator must be offset from the nominal carrier by ω_0, which complicates tuning.

Final Summary: Weaver's method avoids sharp filters and wideband phase shifters but needs four multipliers, two quadrature oscillator pairs and matched filters; imbalance leaves an in-band image, and dc offsets give an audible tone at the band centre.

3. Notes and Remarks

- The phase-shift method's imbalance error lies in the unwanted sideband, outside the channel; Weaver's lies inside it.

7.(g) What is VSB (vestigial sideband)? Why is it used?

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1. Theory

1. The bandwidth–filter dilemma (Core Idea)

DSB wastes bandwidth ($2B$). SSB saves it (B) but needs a filter that cuts sharply exactly at f_c, which is possible only if the message has no energy near dc. Many important signals (television video, digital data) have strong low-frequency content, so neither is ideal.

2. Answer

VSB (vestigial sideband) is a form of amplitude modulation in which one sideband is transmitted almost completely, and only a small part, a **vestige**, of the other sideband is transmitted. It is made by passing a DSB signal through a shaping filter $H_i(f)$ that rolls off **gradually** across f_c.

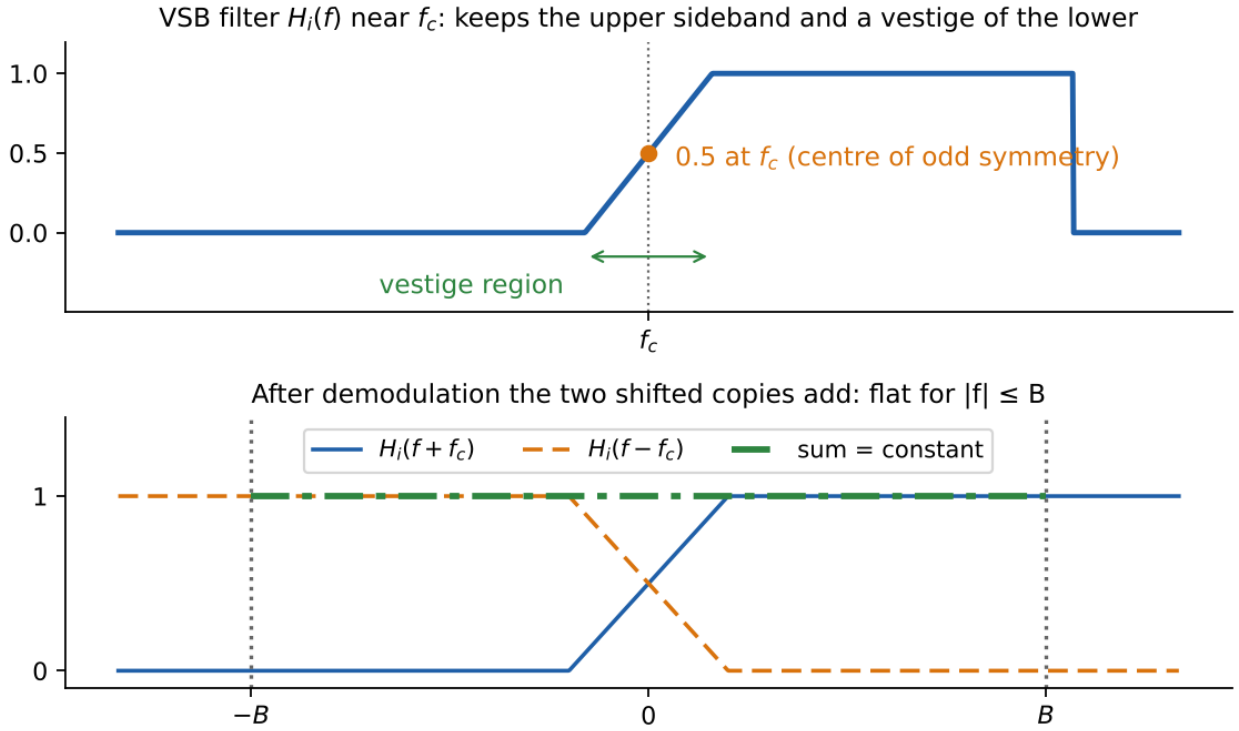


Figure 3: The VSB filter near the carrier (top) and why the vestige makes up for the lost part (bottom); reused from the 2021-22 paper

Reading the figure. Top: the solid blue VSB filter passes the upper sideband fully, rises gradually through the green vestige region around f_c and is 0.5 at f_c (orange dot), so a vestige of the lower sideband gets through. Bottom: after demodulation the two shifted copies of the filter, solid blue and dashed orange, add up to the dash-dot green constant over the message band $|f| \leq B$, so no part of the message is lost.

Why it is used:

1. **Bandwidth close to SSB:** $B_{VSB} = B + f_v$, where f_v is the vestige width, much less than DSB's $2B$.
2. **Realizable filter:** the gentle roll-off around f_c is easy to build, unlike SSB's ideal cut.
3. **Works for signals with low-frequency content:** the frequencies near dc lost in one sideband are made up by the vestige of the other, provided the filter is complementary, $H_i(f_c+f)+H_i(f_c-f) = 1$; coherent detection then gives the message without distortion.
4. **Envelope detection possible** when a carrier is added (VSB+C).

Final Summary: VSB = one sideband plus a vestige of the other; used because it needs little more bandwidth than SSB, uses a practical filter and suits wideband signals with content near dc (TV video, data).

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3. Notes and Remarks

- Lathi, Sec. 4.5. The filter condition is derived in 2(d) of the 2021-22 paper (link), as extra reading.

7.(h) Why is VSB-WC chosen for TV broadcasting?

1. Theory

1. The TV picture signal (Core Idea)

The video signal has a bandwidth of about **4.2 MHz** (NTSC; 5 to 6 MHz in other systems) and, because large areas of a picture have a steady brightness, it has **strong components down to 0 Hz**.

2. Answer

1. **SSB is impossible:** with large energy near dc there is no gap between the sidebands, so no sideband filter can separate them.
2. **DSB wastes too much bandwidth:** DSB would need about 8.4 MHz per channel; with VSB the video needs only about 4.2 MHz plus a vestige of about 1.25 MHz, which fits (with the sound carrier) in a 6 MHz channel.
3. **A practical filter:** VSB's gradual roll-off is realizable.
4. **With carrier (WC), receivers stay cheap:** adding a strong carrier lets every TV set use a simple **envelope detector** instead of a coherent detector with carrier recovery. With a large carrier the envelope is close to $A+m(t)$ and the small distortion caused by the vestige is acceptable.
5. **Broadcast economics:** one transmitter serves millions of receivers, so it pays to spend extra power on the carrier at the transmitter to make every receiver simple and cheap.

Final Summary: VSB-WC is chosen because video has strong low frequencies (no SSB), DSB would double the bandwidth, VSB's filter is practical, and the added carrier allows cheap envelope detection in millions of TV sets.

3. Notes and Remarks

- Lathi, Sec. 4.5 (use of VSB in broadcast television).

7.(i) Derive the narrowband FM ($\beta < 1$) expression.

1. Theory

1. FM in general (Core Idea)

In FM the instantaneous frequency follows the message: $\omega_i(t) = \omega_c + k_f m(t)$. The phase is the integral of the frequency, so

$$\varphi_{FM}(t) = A\cos\left[\omega_c t + k_f \int_{-\infty}^{t} m(\alpha)\,d\alpha\right] = A\cos[\omega_c t + k_f a(t)], \qquad a(t) = \int_{-\infty}^{t} m(\alpha)\,d\alpha$$

2. What narrowband means

Narrowband FM (NBFM) is FM in which the phase deviation stays small: $|k_f a(t)| \ll 1$ (for a tone, $\beta \ll 1$). Then the cosine of a sum can be simplified with small-angle approximations.

2. Answer

Steps 2 and 3 are marked **Basic** (the identity and the series); a student who knows them can skip them.

Step 1: Start from FM

$$\varphi_{FM}(t) = A\cos[\omega_c t + k_f a(t)]$$

Step 2 (Basic): Cosine of a sum

$$\cos(x+y) = \cos x \cos y - \sin x \sin y$$

With $x = \omega_c t$ and $y = k_f a(t)$:

$$\varphi_{FM}(t) = A\left[\cos \omega_c t \cos k_f a(t) - \sin \omega_c t \sin k_f a(t)\right]$$

Step 3 (Basic): Small-angle approximations

From the power series $\cos y = 1 - \frac{y^2}{2!} + \cdots$ and $\sin y = y - \frac{y^3}{3!} + \cdots$, for $|y| \ll 1$:

$$\cos k_f a(t) \approx 1, \qquad \sin k_f a(t) \approx k_f a(t)$$

Step 4: The NBFM expression

$$\boxed{\varphi_{NBFM}(t) \approx A \cos \omega_c t - A k_f a(t) \sin \omega_c t}$$

Step 5: Tone modulation

For $m(t) = A_m \cos \omega_m t$: $k_f a(t) = \dfrac{k_f A_m}{\omega_m} \sin \omega_m t = \beta \sin \omega_m t$, and with $\sin A \sin B = \frac{1}{2}[\cos(A-B) - \cos(A+B)]$:

$$\varphi_{NBFM}(t) \approx A \cos \omega_c t + \frac{A\beta}{2} \cos(\omega_c + \omega_m)t - \frac{A\beta}{2} \cos(\omega_c - \omega_m)t$$

a carrier plus two side frequencies at $f_c \pm f_m$, so the bandwidth is $2B$, as in AM.

Comparison with AM: AM is $A \cos \omega_c t + m(t) \cos \omega_c t$: its sideband term is **in phase** with the carrier. NBFM's sideband term multiplies $\sin \omega_c t$: it is **in quadrature** (90° away), and the lower side frequency has the opposite sign. This is why NBFM can be generated with a DSB-SC (balanced) modulator and a 90° phase shift.

Steps to remember

$A \cos[\omega_c t + k_f a(t)] \to \cos(x+y)$ expansion $\to$ small angle: cos ≈ 1, sin $\approx$ angle $\to A \cos \omega_c t - A k_f a(t) \sin \omega_c t \to$ tone: carrier $\pm$ sidebands $A\beta/2$.

3. Notes and Remarks

- Lathi, Sec. 5.2 (narrowband angle modulation).

7.(j) Draw the NBFM block diagram.

1. Theory

1. Building the NBFM formula (Core Idea)

From 7(i), $\varphi_{NBFM}(t) \approx A \cos \omega_c t - A k_f a(t) \sin \omega_c t$ with $a(t) = \int m\, dt$. Each piece can be made with a standard block: an **integrator** for $a(t)$, a **balanced modulator** (multiplier) for $a(t) \sin \omega_c t$, a **−90° phase shifter** to make $\sin \omega_c t$ from the oscillator's $\cos \omega_c t$, and a **summer** to subtract the product from the carrier.

2. Answer

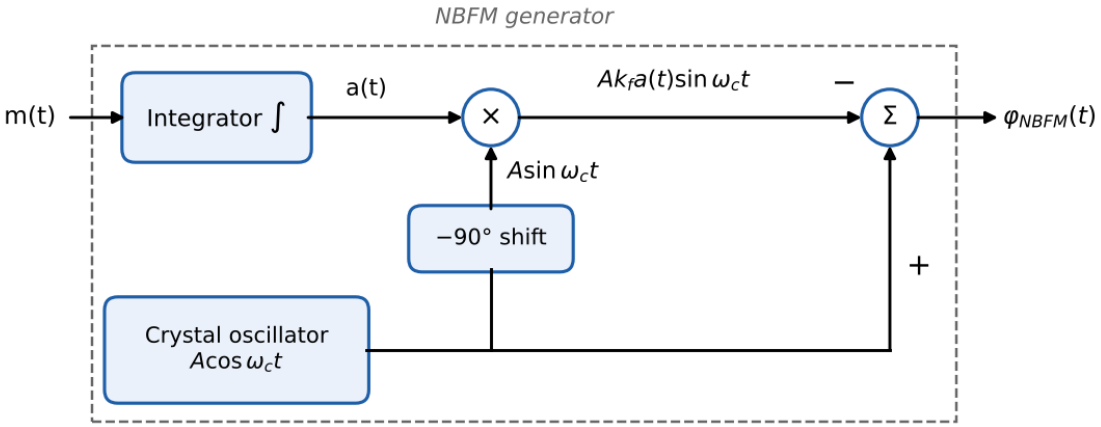


Figure 4: NBFM generator: integrator, balanced modulator, −90° shifter and summer (reused from the 2021-22 paper)

Reading the figure. The message $m(t)$ passes an integrator to give $a(t)$, which is multiplied by $A\sin\omega_c t$ (the crystal oscillator's $A\cos\omega_c t$ shifted by −90°). The product $Ak_f a(t)\sin\omega_c t$ is subtracted from the carrier $A\cos\omega_c t$ in the summer, giving $\varphi_{NBFM}(t)$. The dashed box is the NBFM generator.

- **Integrator:** turns $m(t)$ into $a(t) = \int m\,dt$ (without it the same circuit gives narrowband PM).
- **Crystal oscillator:** gives a very stable carrier $A\cos\omega_c t$.
- **−90° phase shifter:** gives $A\sin\omega_c t$.
- **Balanced modulator:** forms $k_f a(t) A\sin\omega_c t$ (DSB-SC).
- **Summer:** subtracts it from the carrier.

The output is NBFM with a small deviation; in Armstrong's method it is followed by frequency multipliers and a mixer (see 8(a)).

3. Notes and Remarks

- Lathi, Sec. 5.3, Fig. 5.8.

4. Final Concise Answer

(a) FM: better noise immunity, constant envelope, capture effect, fidelity; but much more bandwidth, shorter range, complex equipment, threshold effect.
(b) $\mu = 0.2$; 100.5 and 99.5 kHz; 5 V each; 1 kHz; 2.125 W; 1.96 %.
(c) A large carrier switches the diode pairs, so the output is $m(t)w(t)$ with $w = \pm 1$; the BPF keeps $\frac{4}{\pi} m\cos\omega_c t$; for $m = 0$ the balanced currents cancel and no carrier appears.
(d) AM: $2B$, envelope detector, ≤ 33 % efficient; DSB-SC: $2B$, coherent, 100 %; SSB-SC: B, coherent, 100 %, hardest modulator.
(e) DSB-SC: $m\cos(\Delta\omega t + \delta)$ (attenuation, null, beating); SSB: phase shift by δ, frequency shift by Δf.
(f) Four multipliers, two quadrature oscillator pairs, matched LPFs; imbalance gives an in-band image; dc offset gives a tone at the band centre.
(g) One sideband plus a vestige of the other; bandwidth near SSB, practical filter, works for signals with content near dc.
(h) Video has strong low frequencies (no SSB), DSB doubles the bandwidth, VSB's filter is practical, and the carrier allows cheap envelope detectors.
(i) $\varphi_{NBFM} \approx A\cos\omega_c t - Ak_f a(t)\sin\omega_c t$, $a = \int m\,dt$.
(j) Integrator → balanced modulator with $A\sin\omega_c t$ (oscillator via −90°) → subtracted from $A\cos\omega_c t$.

8. (a) Design an Armstrong indirect FM modulator to generate an FM signal with carrier frequency 97.3MHz and $\Delta f = 20.48kHz$. A NBFM generator of $f_{c1} = 20kHz$ and $\Delta f = 5Hz$ is available. Only frequency doublers can be used as multipliers. Additionally, a local oscillator (LO) with adjustable frequency between 400 and 50-kHz [*sic*: should be "500 kHz"] is readily available for frequency mixing. Marks: 10

   (b) Draw the block diagram of a Differential PSK (DPSK) modulator and demodulator and show how the I_k sequence [1010011100] changes in different parts of the block. Marks: 4

   (c) Two channels, one with a bit rate of 190 kbps and another with a bit rate of 180 kbps, are to be multiplexed using pulse stuffing TDM with no synchronization bits. Answer the following questions:

   a. What is the size of a frame in bits?
   b. What is the frame rate?
   c. What is the duration of a frame?
   d. What is the data rate? Marks: 6

8.(a) Design an Armstrong indirect FM modulator to generate an FM signal with carrier frequency 97.3MHz and $\Delta f = 20.48kHz$. A NBFM generator of $f_{c1} = 20kHz$ and $\Delta f = 5Hz$ is available. Only frequency doublers can be used as multipliers. Additionally, a local oscillator (LO) with adjustable frequency between 400 and 50-kHz [*sic*: should be "500 kHz"] **is readily available for frequency mixing. Marks: 10**

1. Theory

1. Armstrong's indirect method (Core Idea)

A crystal oscillator is very stable but can be frequency-modulated only slightly. So Armstrong's method first makes **narrowband FM** from a crystal oscillator (deviation of a few hertz), then raises the deviation with **frequency multipliers**.

2. Multipliers and mixers

- A **multiplier** by n multiplies *both* the carrier frequency and the deviation by n. k doublers in cascade give $n = 2^k$.
- A **mixer** (a multiplier with a local oscillator, then a bandpass filter) moves the carrier to $f_c \pm f_{LO}$ but leaves the deviation **unchanged**. It corrects the carrier, which the multipliers have pushed too high.

2. Answer

Note on the paper

The paper prints the LO range as "between 400 and 50-kHz"; a range from 400 kHz down to 50 kHz is not meaningful, and the same problem (Lathi's, also asked in the 2019-20 final) has 400 to 500 kHz, so 500 kHz is used.

Given: NBFM $f_{c1} = 20$ kHz, $\Delta f_1 = 5$ Hz; output $f_c = 97.3$ MHz, $\Delta f = 20.48$ kHz; doublers only; LO 400 to 500 kHz.

Step 1: Total multiplication

$$n_1 n_2 = \frac{\Delta f}{\Delta f_1} = \frac{20\,480}{5} = 4096 = 2^{12}$$

So **12 doublers** in all, split as $n_1 = 2^a$ before the mixer and $n_2 = 2^{12-a}$ after it.

Step 2: Carriers around the mixer

- after the first multiplier: $f_2 = 20$ kHz $\times\, n_1$
- needed before the second multiplier: $f_3 = 97.3$ MHz$/n_2 = 97.3$ MHz $\times\, n_1/4096 = 23\,754.88\, n_1$ Hz

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- the mixer must give f_3 from f_2 with $400 \le f_{LO} \le 500$ kHz.

Step 3: Find the split

Sum frequency: $f_{LO} = f_3 - f_2 = (23\,754.88 - 20\,000)\, n_1 = 3754.88\, n_1$ Hz. For $400 \le f_{LO} \le 500$ kHz:

$$106.5 \le n_1 \le 133.2 \quad \Rightarrow \quad n_1 = 128 = 2^7 \text{ (the only power of two)}$$

(Difference $f_3 = f_{LO} - f_2$ needs $f_{LO} = 43\,754.88\, n_1$, in range only for $9.1 \le n_1 \le 11.4$: no power of two; and $f_3 = f_2 - f_{LO}$ is impossible because $f_3 > f_2$.)
So $n_1 = 128$ (7 doublers) and $n_2 = 4096/128 = 32$ (5 doublers).

Step 4: The frequencies

- After $\times 128$: $f_2 = 128 \times 20 = 2560$ kHz, $\Delta f_2 = 128 \times 5 = 640$ Hz
- Needed after the mixer: $f_3 = 97.3$ MHz$/32 = 3040.625$ kHz (deviation still 640 Hz)
- LO: $f_{LO} = 3040.625 - 2560 = 480.625$ kHz, within 400 to 500 kHz; the BPF keeps the **sum**
- After $\times 32$: $f_c = 32 \times 3040.625$ kHz $= 97.3$ MHz, $\Delta f = 32 \times 640 = 20\,480$ Hz ✔

$$\boxed{n_1 = 128 \text{ (7 doublers)}, \quad f_{LO} = 480.625 \text{ kHz (sum)}, \quad n_2 = 32 \text{ (5 doublers)}}$$

Step 5: The block diagram

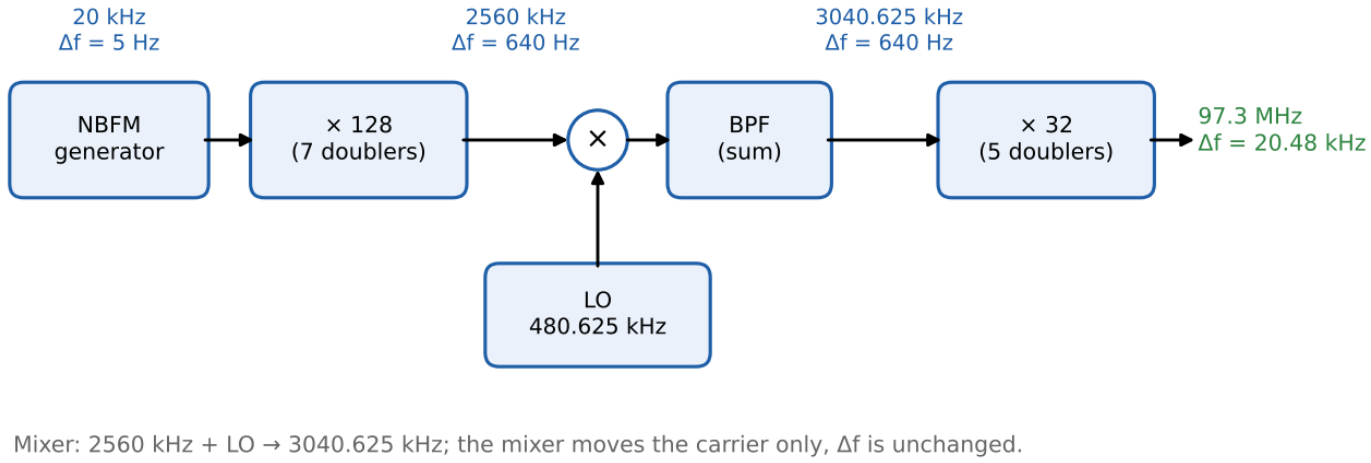


Figure 5: Armstrong indirect FM modulator for 97.3 MHz with 20.48 kHz deviation

Reading the figure. The NBFM generator (20 kHz, 5 Hz) feeds the $\times 128$ multiplier (7 doublers): 2560 kHz, 640 Hz. The mixer with the 480.625 kHz LO and the BPF keep the sum, 3040.625 kHz, still 640 Hz. The $\times 32$ multiplier (5 doublers) gives 97.3 MHz with 20.48 kHz deviation (green).
Final Answer: $\times 128$ (7 doublers) $\rightarrow$ mixer, $f_{LO} = 480.625$ kHz, sum 3040.625 kHz $\rightarrow$ $\times 32$ (5 doublers): 97.3 MHz, $\Delta f = 20.48$ kHz.

Steps to remember

$n_1 n_2 = \Delta f / \Delta f_1 \rightarrow f_2 = n_1 f_{c1}$, $f_3 = f_c / n_2 \rightarrow f_{LO} = |f_3 - f_2|$ must be in the LO range $\rightarrow$ choose the power of two that fits $\rightarrow$ check f_c and Δf.

3. Notes and Remarks

- The 2019-20 final asked the same design with $\Delta f = 10.24$ kHz; there $n_1 = 16$, $f_{LO} = 440.156$ kHz, $n_2 = 128$ (link). Doubling the deviation moves three doublers ahead of the mixer.

8.(b) Draw the block diagram of a Differential PSK (DPSK) modulator and demodulator and show how the I_k sequence [1010011100] changes in different parts of the block. Marks: 4

1. Theory

1. Why differential? (Core Idea)

PSK sends a 1 as $+A\cos\omega_c t$ and a 0 as $-A\cos\omega_c t$; detecting it needs a local carrier with the exact phase (coherent detection). **DPSK** codes the data in the **change** of phase between successive bits, so the receiver can use the previous received bit as its phase reference and needs no local carrier.

2. Differential encoding

$$q_k = I_k \oplus q_{k-1}$$

(XOR, Lathi's convention): a data 1 flips q, a data 0 keeps it. The encoded bit is sent by PSK with $a_k = +1$ for $q_k = 1$ and -1 for $q_k = 0$.

2. Answer

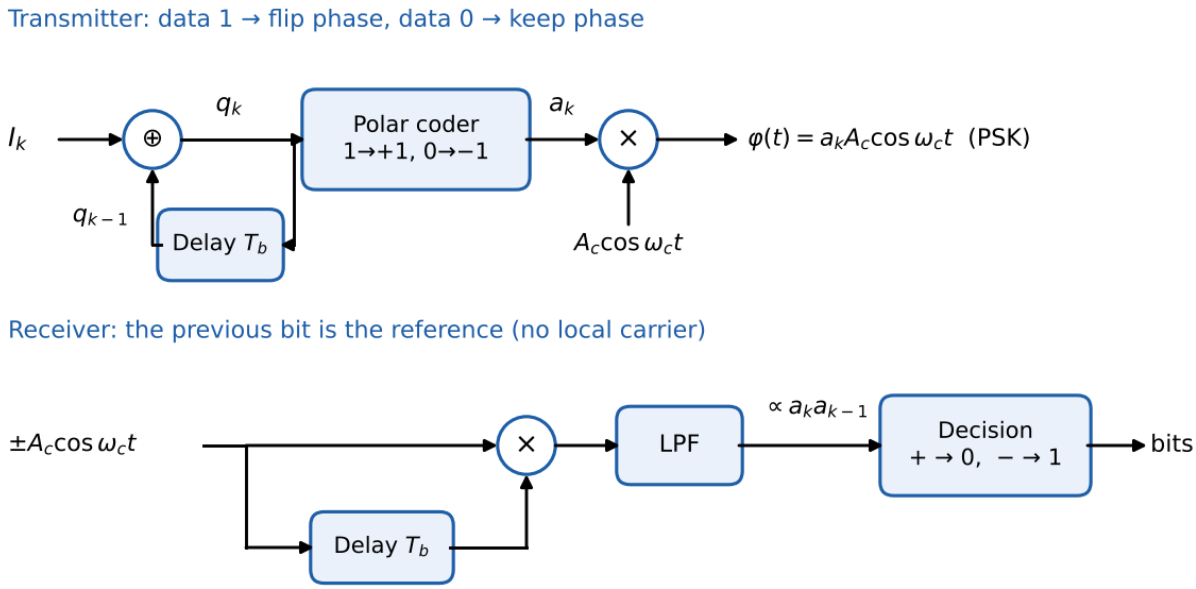


Figure 6: DPSK modulator (XOR with the delayed encoded bit, polar coder, carrier multiplier) and demodulator (delay-and-multiply, LPF, decision); reused from the 2021-22 paper

Reading the figure. Transmitter: I_k and the previous encoded bit q_{k-1} (from the delay T_b) go into the XOR; q_k is made polar ($a_k = \pm 1$) and multiplies $A_c\cos\omega_c t$. Receiver: the signal is multiplied by itself delayed by T_b, low-pass filtered to a value proportional to $a_k a_{k-1}$, and the decision gives 0 for positive, 1 for negative.

Assumption: reference bit $q_0 = 0$.

k	0	1	2	3	4	5	6	7	8	9	10
Input I_k		1	0	1	0	0	1	1	1	0	0
Delayed q_{k-1}		0	1	1	0	0	0	1	0	1	1
XOR output q_k	0 (ref)	1	1	0	0	0	1	0	1	1	1
Polar a_k	-1	$+1$	$+1$	-1	-1	-1	$+1$	-1	$+1$	$+1$	$+1$
Carrier phase	π	0	0	π	π	π	0	π	0	0	0

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k	0	1	2	3	4	5	6	7	8	9	10
Receiver LPF, sign of $a_k a_{k-1}$		−	+	−	+	+	−	−	−	+	+
Decision (output)		1	0	1	0	0	1	1	1	0	0

Table 3: Summary

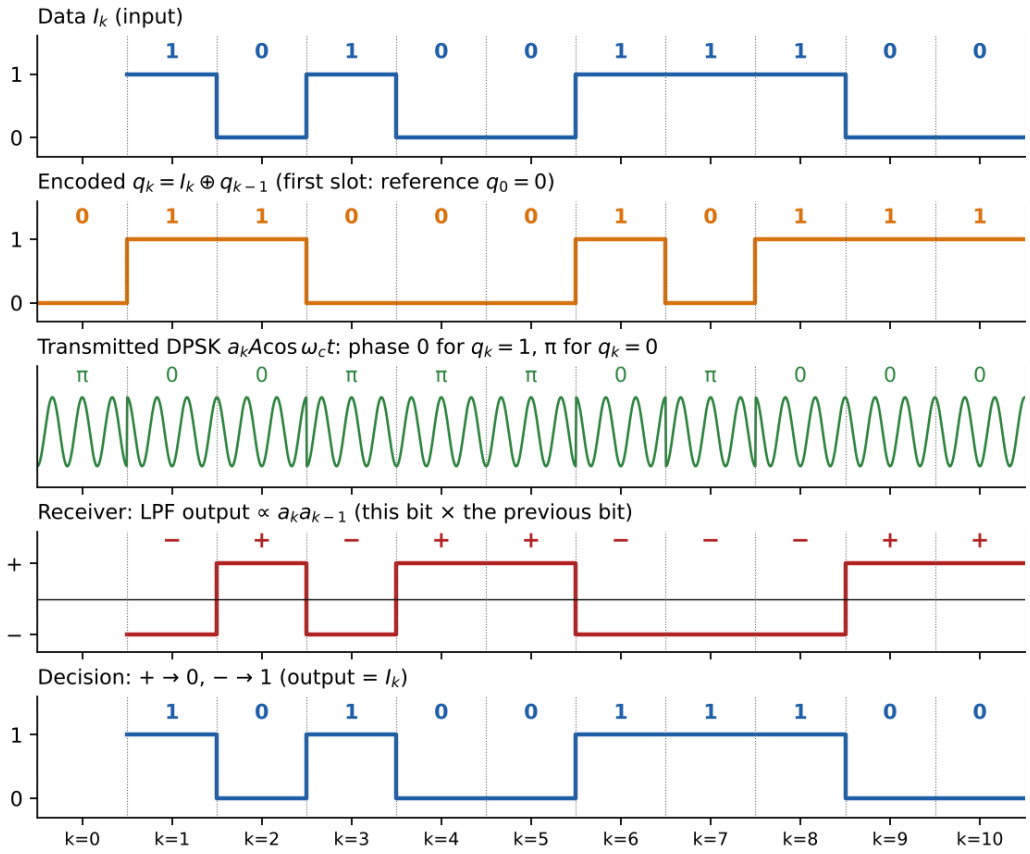


Figure 7: How 1010011100 changes through the blocks: data, encoded bits, transmitted DPSK, receiver LPF output and the decision

Reading the figure. Row by row, with bit slots $k = 0$ to 10 marked by dotted lines: the blue data I_k; the orange encoded bits q_k, which flip at every data 1; the green transmitted carrier, whose phase (0 or π, written above) follows q_k; the red receiver output, the sign of this bit times the previous bit; and the blue decision, which equals the input data.

Step 1: Encoder

Each q_k is the XOR of I_k with the previous q: for example $q_1 = 1 \oplus 0 = 1$, $q_2 = 0 \oplus 1 = 1$, $q_3 = 1 \oplus 1 = 0$.

Step 2: Modulator

q_k becomes $a_k = \pm 1$ and the carrier is sent as $a_k A \cos \omega_c t$: phase 0 for $q_k = 1$, π for $q_k = 0$.

Step 3: Demodulator

The product of the received bit and the previous one, after the LPF, is $\frac{1}{2} a_k a_{k-1} A^2$ (from $\cos^2 x = \frac{1}{2}(1 + \cos 2x)$): positive when the phase did not change (data 0), negative when it changed (data 1).

Final Answer: encoded q_k = 1 1 0 0 0 1 0 1 1 1 (reference 0); phases 0 0 π π π 0 π 0 0 0; receiver signs − + − + + − − − + +; output 1010011100, equal to the input.

3. Notes and Remarks

- With the reference $q_0 = 1$ every q_k and phase would be inverted, but the received signs and the output would be the same: only phase **changes** matter.
- The same sequence is solved as 6(d) of the 2020-21 paper.

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8.(c) Two channels, one with a bit rate of 190 kbps and another with a bit rate of 180 kbps, are to be multiplexed using pulse stuffing TDM with no synchronization bits. Answer the following questions: a. What is the size of a frame in bits? b. What is the frame rate? c. What is the duration of a frame? d. What is the data rate? Marks: 6

1. Theory

1. Pulse stuffing (Core Idea)

Synchronous TDM gives each channel one slot per frame, so all channels must run at the same rate. When the rates differ and are not integer multiples of each other, **pulse stuffing** adds dummy bits to the slower channel until it runs at the rate of the faster one. The receiver knows where the dummy bits are and discards them.

2. Answer

Given: channel A 190 kbps, channel B 180 kbps; no synchronization bits. **Assumption:** each slot carries 1 bit.

Step 1: Stuffing

Channel B is raised to 190 kbps by adding $190 - 180 = 10$ kbps of dummy bits. Both channels now deliver 190 000 bits/s.

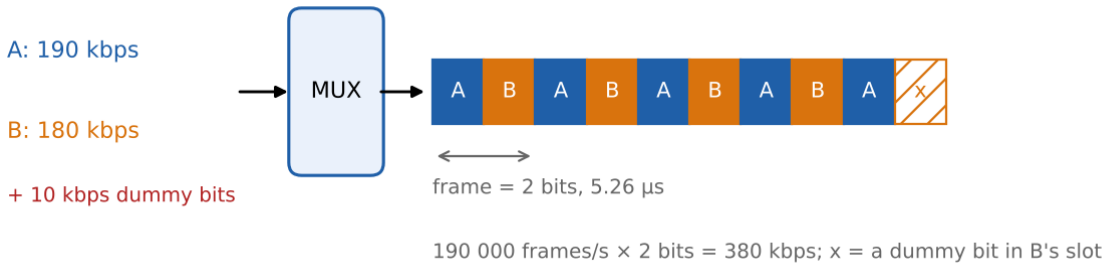


Figure 8: Pulse stuffing: channel B gets 10 kbps of dummy bits (x) so that both run at 190 kbps; each frame has one bit from each

Reading the figure. A (blue) at 190 kbps and B (orange) at 180 kbps plus 10 kbps of dummy bits enter the MUX. The output alternates one A bit and one B bit; now and then B's slot holds a dummy bit (white, hatched x). Each frame is 2 bits long and lasts 5.26 μs.

Step 2: (a) Frame size

One bit from each of the two channels:

$$\text{frame} = 2 \text{ bits}$$

Step 3: (b) Frame rate

Each frame takes one bit from each channel, and each channel (after stuffing) gives 190 000 bits/s:

$$\text{frame rate} = 190\,000 \text{ frames/s}$$

Step 4: (c) Frame duration

$$T_f = \frac{1}{190\,000} = 5.26 \ \mu\text{s}$$

Step 5: (d) Data rate

$$R = 190\,000 \times 2 = 380\,000 \text{ bits/s} = 380 \text{ kbps}$$

$$\boxed{2 \text{ bits},\ 190\,000 \text{ frames/s},\ 5.26\ \mu\text{s},\ 380 \text{ kbps}}$$

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Final Answer: 2 bits per frame; 190 000 frames/s; 5.26 μs per frame; 380 kbps (of which 10 kbps are stuffed bits).

3. Notes and Remarks

- Forouzan, Sec. 6.2 (pulse stuffing). The link carries $380 - 370 = 10$ kbps of dummy bits.

4. Final Concise Answer

(a) $n_1 n_2 = 4096$: $\times 128$ (7 doublers) $\to$ 2560 kHz, 640 Hz $\to$ mixer, $f_{LO} = 480.625$ kHz (sum) $\to$ 3040.625 kHz $\to \times 32$ (5 doublers) $\to$ 97.3 MHz, 20.48 kHz.

(b) $q_k = I_k \oplus q_{k-1}$: 1 1 0 0 0 1 0 1 1 1 (ref 0); phases 0 0 π π π 0 π 0 0 0; receiver signs $-\ +\ -\ +\ +\ -\ -\ -\ +\ +$; output 1010011100.

(c) 2 bits; 190 000 frames/s; 5.26 μs; 380 kbps.

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FORMULA SHEET

EEE 0714 3229

Communication Engineering I

Formula Sheet

Session 2022–23 term tests *Every formula behind Term Tests 2 and 3; the final examination will be added*

TOPICS

1 Amplitude modulation — 12 formulas

2 Angle modulation — 7 formulas

3 Digital modulation and multiplexing — 4 formulas

Each topic is one table: **Title** · **Equation** · **Notation** · **Reference**.

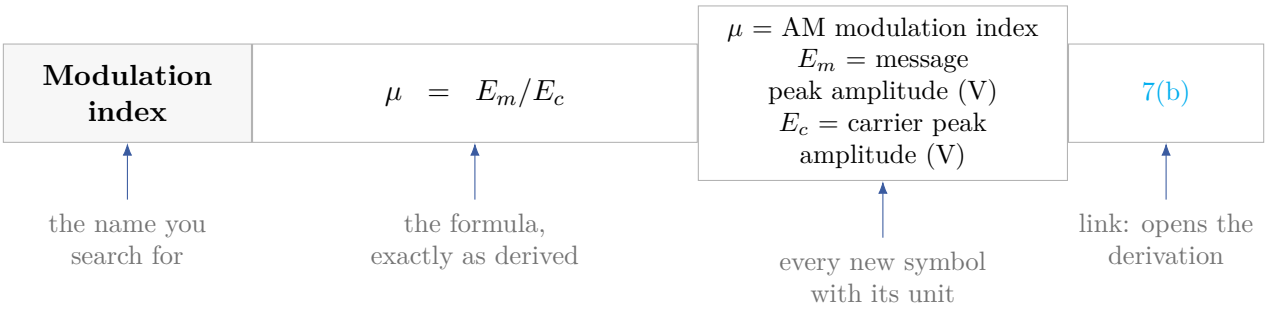
Prepared by Nur Mohammad

November 2026

BACK OF THE COVER

Before you use the sheet

How to read a row



In the exam

- Write the formula first, then put the numbers in: the formula alone often earns a mark.
- Read the symbols in the Notation column before you substitute, and check the units.
- Not sure where a formula comes from? Its Reference opens the derivation.
- The **Key formulas** page at the end gathers the rows marked ★.
- Lost a symbol? The **Symbol index** at the end links to the row where it first appears.

Contents

1 Amplitude modulation 4

2 Angle modulation 5

3 Digital modulation and multiplexing 6

4 Key formulas 7

5 Symbol index 8

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Conventions (Lathi and Ding, 4th ed.): $\omega = 2\pi f$; $m(t)$ is the message, $m_h(t)$ its Hilbert transform and B its bandwidth (Hz); μ is the AM modulation index and β the FM one. References point to the solutions of the 2022–23 term tests.

Columns: Title · Equation · Notation · Ref. (reference).

1. Amplitude modulation:

Title	Equation	Notation	Ref.
Modulation index	$\mu = E_m/E_c$	μ = AM modulation index E_m = message peak amplitude (V) E_c = carrier peak amplitude (V)	7(b)
Side frequencies	$f_c \pm f_m$, amplitude $\mu E_c/2$	f_c = carrier frequency (Hz) f_m = message (tone) frequency (Hz)	7(b)
Carrier power	$P_c = \dfrac{E_c^2}{2R}$	P_c = carrier power (W) R = load resistance (Ω)	7(b)
★ **Total AM power**	$P_t = P_c\left(1 + \dfrac{\mu^2}{2}\right)$	P_t = total transmitted power (W)	7(b)
★ **Efficiency**	$\eta = \dfrac{\mu^2}{2+\mu^2}$	η = power efficiency (sideband power / total power) at most 33.3 % ($\mu = 1$)	7(b)
Bandwidth	AM, DSB-SC: $2B$; SSB: B	B = message bandwidth (Hz)	7(d)
Square wave (± 1)	$w(t) = \dfrac{4}{\pi}\left[\cos\omega_c t - \frac{1}{3}\cos 3\omega_c t + \cdots\right]$	$w(t)$ = switching square wave, ± 1 t = time (s) ω_c = carrier angular frequency (rad/s) no dc term	7(c)
★ **Ring modulator output**	$\dfrac{4}{\pi} m(t)\cos\omega_c t$ after the BPF	$m(t)$ = message signal (V) $m = 0$ gives 0 (carrier suppressed)	7(c)
DSB-SC with LO errors	$e_o = m(t)\cos(\Delta\omega t + \delta)$	e_o = demodulator output (V) $\Delta\omega$ = LO frequency error (rad/s) δ = LO phase error (rad)	7(e)
★ **SSB with LO errors**	$e_o = m\cos(\Delta\omega t + \delta) - m_h \sin(\Delta\omega t + \delta)$	$m_h(t)$ = Hilbert transform of $m(t)$ a phase error shifts each component's phase by δ; a frequency error shifts its frequency by $\Delta\omega/2\pi$	7(e)
VSB filter condition	$H_i(f_c + f) + H_i(f_c - f) = 1$	$H_i(f)$ = transfer function of the VSB filter f = frequency measured from the carrier (Hz) holds over the message band	7(g)
VSB bandwidth	$B + f_v$	f_v = vestige width (Hz)	7(g)

Table 1: Amplitude modulation

2. Angle modulation:

Title	Equation	Notation	Ref.
FM signal	$A\cos[\omega_c t + k_f a(t)]$, $a(t) = \int m\,dt$	A = carrier amplitude (V) ω_c = carrier angular frequency (rad/s) t = time (s) k_f = frequency sensitivity (rad/s per V) $m(t)$ = message signal (V) $a(t)$ = integral of the message (V s)	7(i)
★ **NBFM**	$\varphi_{NBFM} \approx A\cos\omega_c t - Ak_f a(t)\sin\omega_c t$	φ_{NBFM} = narrowband FM signal valid for $\lvert k_f a(t)\rvert \ll 1$	7(i)
NBFM, tone	$A\cos\omega_c t + \frac{A\beta}{2}\cos(\omega_c+\omega_m)t - \frac{A\beta}{2}\cos(\omega_c-\omega_m)t$	β = FM modulation index ω_m = tone angular frequency (rad/s) bandwidth: twice the tone frequency	7(i)
Carson's rule	$B_T = 2(\Delta f + B)$	B_T = FM transmission bandwidth (Hz) Δf = peak frequency deviation (Hz) B = message bandwidth (Hz) FM needs far more than AM	7(a)
Multiplier	$f_c \to nf_c$, $\Delta f \to n\Delta f$	f_c = carrier frequency (Hz) n = multiplication factor $n = 2^k$ for k doublers	8(a)
Mixer	$f_c \to f_c \pm f_{LO}$, Δf unchanged	f_{LO} = local-oscillator frequency (Hz)	8(a)
★ **Armstrong design**	$n_1 n_2 = \dfrac{\Delta f}{\Delta f_1}$, $f_{LO} = \left\lvert \dfrac{f_c}{n_2} - n_1 f_{c1}\right\rvert$	n_1, n_2 = multiplication factors before and after the mixer Δf_1 = deviation of the NBFM generator (Hz) f_{c1} = carrier of the NBFM generator (Hz) $n_1 = 128$, $n_2 = 32$, $f_{LO} = 480.625$ kHz	8(a)

Table 2: Angle modulation

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3. Digital modulation and multiplexing:

Title	Equation	Notation	Ref.
★ **DPSK encoder**	$q_k = I_k \oplus q_{k-1}$	q_k = encoded bit k q_{k-1} = previous encoded bit I_k = data bit k data 1 flips the phase	8(b)
DPSK receiver	LPF output $= \frac{1}{2} a_k a_{k-1} A^2$	a_k = sign (± 1) of the received pulse k a_{k-1} = sign of the previous pulse A = carrier amplitude (V) $+ \rightarrow 0$, $- \rightarrow 1$	8(b)
Pulse stuffing	slow channel padded to the fast rate	dummy bits: $190 - 180 = 10$ kbps	8(c)
★ **TDM frame**	frame rate = channel rate / bits per slot; R = frame rate $\times$ bits per frame	R = output bit rate (bps) 190 000 frames/s, 380 kbps	8(c)

Table 3: Digital modulation and multiplexing

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4. Key formulas:

The formulas marked ★ in the sheet, on one page. Each title links back to its row.

Title	Equation
Total AM power	$P_t = P_c\left(1 + \frac{\mu^2}{2}\right)$
Efficiency	$\eta = \frac{\mu^2}{2+\mu^2}$
Ring modulator output	$\frac{4}{\pi} m(t)\cos\omega_c t$ after the BPF
SSB with LO errors	$e_o = m\cos(\Delta\omega t + \delta) - m_h \sin(\Delta\omega t + \delta)$
NBFM	$\varphi_{NBFM} \approx A\cos\omega_c t - Ak_f a(t)\sin\omega_c t$
Armstrong design	$n_1 n_2 = \frac{\Delta f}{\Delta f_1}$, $f_{LO} = \left\lvert \frac{f_c}{n_2} - n_1 f_{c1} \right\rvert$
DPSK encoder	$q_k = I_k \oplus q_{k-1}$
TDM frame	frame rate = channel rate / bits per slot; R = frame rate × bits per frame

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5. Symbol index:

Every symbol the sheet defines, in the order it first appears. The last column links to that row.

Symbol	Meaning	First used in
μ	AM modulation index	Modulation index
E_m	message peak amplitude (V)	Modulation index
E_c	carrier peak amplitude (V)	Modulation index
f_c	carrier frequency (Hz)	Side frequencies
f_m	message (tone) frequency (Hz)	Side frequencies
P_c	carrier power (W)	Carrier power
R	load resistance (Ω)	Carrier power
P_t	total transmitted power (W)	Total AM power
η	power efficiency (sideband power / total power)	Efficiency
B	message bandwidth (Hz)	Bandwidth
$w(t)$	switching square wave, ± 1	Square wave (± 1)
t	time (s)	Square wave (± 1)
ω_c	carrier angular frequency (rad/s)	Square wave (± 1)
$m(t)$	message signal (V)	Ring modulator output
e_o	demodulator output (V)	DSB-SC with LO errors
$\Delta\omega$	LO frequency error (rad/s)	DSB-SC with LO errors
δ	LO phase error (rad)	DSB-SC with LO errors
$m_h(t)$	Hilbert transform of $m(t)$	SSB with LO errors
$H_i(f)$	transfer function of the VSB filter	VSB filter condition
f	frequency measured from the carrier (Hz)	VSB filter condition
f_v	vestige width (Hz)	VSB bandwidth
A	carrier amplitude (V)	FM signal
k_f	frequency sensitivity (rad/s per V)	FM signal
$a(t)$	integral of the message (V s)	FM signal
φ_{NBFM}	narrowband FM signal	NBFM
β	FM modulation index	NBFM, tone
ω_m	tone angular frequency (rad/s)	NBFM, tone
B_T	FM transmission bandwidth (Hz)	Carson's rule
Δf	peak frequency deviation (Hz)	Carson's rule
n	multiplication factor	Multiplier
f_{LO}	local-oscillator frequency (Hz)	Mixer
n_1, n_2	multiplication factors before and after the mixer	Armstrong design
Δf_1	deviation of the NBFM generator (Hz)	Armstrong design
f_{c1}	carrier of the NBFM generator (Hz)	Armstrong design
q_k	encoded bit k	DPSK encoder
q_{k-1}	previous encoded bit	DPSK encoder

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Symbol	Meaning	First used in
I_k	data bit k	DPSK encoder
a_k	sign (± 1) of the received pulse k	DPSK receiver
a_{k-1}	sign of the previous pulse	DPSK receiver
R	output bit rate (bps)	TDM frame

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